	Start	Ann. Geo. Mean Return (%)
US 10Y Bond	2000-02	3.60
EU 10Y Bond	2000-02	2.96
US Equity	2000-02	8.39
EU Equity	2000-08	4.75
US HY Credit	2007-05	4.92
EU HY Credit	2008-02	0.46
Commodities	2006-03	2.14

Table: Annualized Geometric Mean Total Returns (since 2000)

Data sources and coverage:

Asset	Source	Instrument	Coverage
US 10Y Bond	FRED `DGS10`	Federal Reserve H.15 constant-maturity Treasury yield; total returns computed via bond math: $r_t \approx y_{t-1}/12 - D_{mod} \cdot \Delta y_t$	Jan 2000 – present
EU 10Y Bond	FRED `IRLTLT01DEM156N`	German 10Y Bund yield from OECD; Germany is the standard euro-area risk-free benchmark. Returns computed identically to US bonds	Jan 2000 – present
US Equity	Yahoo Finance `^SP500TR`	S&P 500 Total Return Index, a **cash index** with dividends reinvested; preferred over the SPY ETF which has tracking error and expense drag	Jan 2000 – present
EU Equity	Yahoo Finance `EZU`	iShares MSCI EMU ETF (adjusted close); tracks MSCI EMU Index covering large/mid-cap across 10 eurozone countries. ETF proxy (no freely available EUR equity cash total return index)	Aug 2000 – present
US HY Credit	Yahoo Finance `HYG`	iShares iBoxx $ High Yield Corporate Bond ETF; tracks Markit iBoxx USD Liquid High Yield Index. ETF proxy; FRED's ICE BofA HY series (`BAMLHYH0A0HYM2TRIV`) is now restricted to the last ~3 years of history only	Apr 2007 – present
EU HY Credit	Yahoo Finance `EXHA.DE`	iShares Euro High Yield Corp Bond UCITS ETF (Xetra, EUR-denominated); tracks Markit iBoxx EUR Liquid High Yield Index. ETF proxy (no Euro HY total return index is freely available)	Jan 2008 – present
Commodities	Yahoo Finance `DBC`	Invesco DB Commodity Index Tracking Fund; tracks the DBIQ Optimum Yield Diversified Commodity Index, **futures-based** across energy, metals, and agriculture. Appropriate since macro investors access commodities via futures; no liquid cash commodity instrument exists	Feb 2006 – present

Table: **Data sources and coverage**

	Ann. Geo. Mean (%)	Ann. Volatility (%)	Return / Vol
US 10Y Bond	3.60	7.60	0.474
EU 10Y Bond	2.96	5.09	0.581
US Equity	8.39	15.19	0.552
EU Equity	4.75	21.06	0.225
US HY Credit	4.92	10.28	0.478
EU HY Credit	0.46	3.73	0.124
Commodities	2.14	18.54	0.115

Table: Risk and Return Summary

	Total Return (%)	Carry / Income (%)	Price / Capital (%)
US 10Y Bond	3.60	3.36	0.24
US Equity	8.39	1.87	6.41

Table: US 10Y Bond and US Equity Comparison

	Total Vol (%)	Carry Vol (%)	Price Vol (%)	Corr(Carry,Price)
US 10Y Bond	7.60	0.37	7.54	0.144
US Equity	15.19	0.19	15.18	0.043

Table: Volatility Decomposition of US 10Y Bond and US Equity

Q7(a): HY vs Equity vs Bonds — Full Risk Profile

	Ann. Geo. Mean (%)	Ann. Vol (%)	Return/Vol	Max Drawdown (%)	Skewness	Excess Kurtosis
US HY Credit	4.92	10.28	0.478	-30.24	0.296	7.602
US Equity	8.39	15.19	0.552	-50.95	-0.489	0.792
US 10Y Bond	3.60	7.60	0.474	-27.29	-0.066	0.891

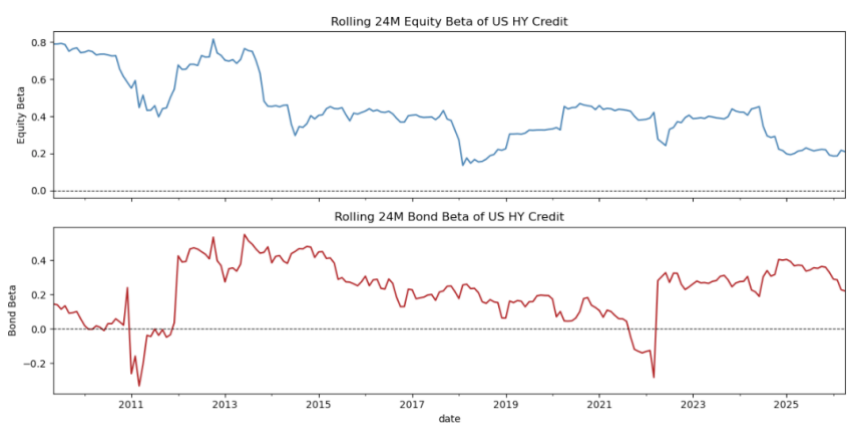
Table : HY vs Equity vs Bonds — Full Risk Profile

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                     HY   R-squared:                       0.556
Model:                            OLS   Adj. R-squared:                  0.552
Method:                 Least Squares   F-statistic:                     141.0
Date:                Sun, 19 Apr 2026   Prob (F-statistic):           2.04e-40
Time:                        14:58:07   Log-Likelihood:                 571.64
No. Observations:                 228   AIC:                            -1137.
Df Residuals:                     225   BIC:                            -1127.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.0006      0.001     -0.451      0.653      -0.003       0.002
Equity         0.4926      0.029     16.754      0.000       0.435       0.551
Bond           0.1543      0.060      2.557      0.011       0.035       0.273
==============================================================================
Omnibus:                      160.226   Durbin-Watson:                   2.225
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             3152.894
Skew:                           2.391   Prob(JB):                         0.00
Kurtosis:                      20.579   Cond. No.                         46.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

R-squared: 0.556 — HY credit is 55.6% explained by equity + bond moves
Equity beta: 0.493
Bond beta:   0.154
```

Table: Regress HY returns on equity + bond returns to estimate implicit exposures



Plot: Rolling 24M beta on equity/bond

	US 10Y Bond (DGS10)	EU 10Y Bond (DE Bund, IRLTLT01DEM156N)	US Equity (^SP500TR)	EU Equity (EZU)	US HY Credit (HYG)	EU HY Credit (EXHA.DE)	Commodities (DBC)
2008-02-01	3.62	3.952857	2194.429932	32.151608	31.968224	113.970909	26.559078
2008-04-01	3.57	4.040000	2163.379883	33.126442	31.789108	114.117302	29.605341
2008-07-01	4.01	4.490435	2039.660034	29.700212	31.329014	107.332596	37.928463
2008-08-01	3.97	4.202381	2003.160034	28.848827	31.406315	108.804886	33.250870
2008-10-01	3.77	3.883478	1853.260010	24.108061	29.148830	111.982025	27.549530

Table: Sample Data From Each Time Series